Upper-half-plane radiative mode exclusion for subextremal Kerr–Newman black holes

Abstract. We prove upper-half-plane radiative mode exclusion for the linearized Einstein–Maxwell equations on every strictly subextremal, asymptotically flat Kerr–Newman black hole. For each fixed azimuthal number and frequency with positive imaginary part, outgoing modes satisfying the regular axis and future-horizon conditions have vanishing gravitational and electromagnetic radiative data. For nonzero charge, the proof treats the full coupled first-order radiative PDE system, using complex angular generalized-root completeness, a division-free radial reduction, a distributional horizon extension, and a coercive transformed energy identity. The Kerr case is closed by a full-field spectral reduction and a spin-reversal lemma that covers exceptional horizon frequencies without assuming a nonzero leading horizon coefficient. The result concerns radiative modes and does not assert a decay theorem or a classification of nonradiative gauge and parameter perturbations.

1 Introduction and theorem

The separation of the coupled gravitational and electromagnetic radiative perturbation equations on subextremal Kerr–Newman backgrounds [H] makes it possible to study the non-real mode problem directly. The real-frequency exclusion established in [H] does not by itself exclude frequencies with positive imaginary part. Here we prove that exclusion for the radiative Einstein–Maxwell variables in the complete strictly subextremal parameter domain. The zero-charge Kerr case is treated separately, using the negative-spin version of the classical separated Kerr result [W, AMPW] and an explicit spin-reversal argument that includes exceptional horizon frequencies. Rigorous linear stability is known in the restricted weak-charge and slow-rotation regime [He]; earlier numerical studies of the coupled system [DGS] and calculations of separated Kerr–Newman quasinormal modes [DCH] provide complementary evidence in broader parameter ranges. Recent work on radiative amplitudes and algebraic separation [M, Z] supplies further context. We use geometrized units throughout.

Fix real parameters

$$M > 0, \qquad a^2 + Q^2 < M^2,$$

an integer m, and a frequency $\omega \in \mathbb{C}$ with $\gamma := \operatorname{Im}\omega > 0$. Time and azimuthal dependence is $e^{-i\omega t + im\phi}$.

Theorem (upper-half-plane radiative mode exclusion). Every outgoing smooth linearized Einstein–Maxwell perturbation on this asymptotically flat Kerr–Newman exterior has vanishing gravitational and electromagnetic radiative data.

For $Q \neq 0$, the stronger sufficient assertion proved below is that both entire four-component radiative first-order PDE systems in Hintz [H], with their prescribed axis, future-horizon and infinity conditions, have only the zero solution. Their equivalence to the transformed equations used here is algebraic and valid for complex frequency. For $Q = 0$, the Kerr section below combines the classical separated Kerr mode-stability theorem in its negative-spin form [W, AMPW] with a spin-reversal lemma and an explicit full-PDE coverage argument.

Put

$$\Delta = r^2 - 2Mr + a^2 + Q^2, \quad r_\pm = M \pm \delta, \quad \delta = \sqrt{M^2 - a^2 - Q^2} > 0,$$

$$K = -(r^2 + a^2)\omega + am, \qquad H(r) = \exp\left(i \int_{2r_+}^{r} K(s)/\Delta(s)\, ds \right).$$

Future-horizon regularity means that $H^{-1}\Delta^b \psi_b$, $H^{-1}\Delta^2 k$, $H^{-1}\Delta s$, $H^{-1}\Delta^{-2} n$, and $H^{-1}\Delta^{-1}\ell$ are smooth up to $r = r_+$, using the variables of the relevant spin system. This is smoothness in the regular ingoing tetrad. At infinity, H times each original radiative variable has polynomial conormal bounds, meaning that every fixed collection of $r\partial_r$ and angular derivatives has a polynomial bound. The physical outgoing branch decays exponentially because $\gamma > 0$. Axis smoothness means smoothness of the relevant spin-weighted sections.

The theorem concerns radiative mode stability. Classification of nonradiating perturbations as gauge and parameter variations, decay, nonlinear stability, extremal backgrounds, and bounds uniform to extremality are outside this claim.

2 Source system and transformations

2.1 Constants and the regular PDE

Assume $Q \neq 0$, and define

$$d = M^2 - Q^2, \quad \eta = \frac{2Q^2}{d}, \quad h = \eta + 2 = \frac{2M^2}{d}, \quad \kappa = \eta + 1 = \frac{M^2 + Q^2}{d},$$

$$c = \frac{2MQ}{d}, \qquad \beta = \frac{3M}{2Q}, \qquad \xi_0 = \eta\omega, \qquad \widetilde{K} = K - \xi_0\Delta.$$

Then

$$\kappa^2 - c^2 = 1, \qquad \beta c = \frac{3h}{2}. \tag{1}$$

Let $\sigma \in \{+1, -1\}$ distinguish the positive and negative spin systems. The exact transformed equation [H, equations (7)–(8)] is

$$\sqrt{\Delta}\,\partial_r Z = (\mathcal{A}_\sigma(\omega) + R_0\mathcal{I} + R_1\mathcal{J})Z, \tag{2}$$

where

$$R_0 = \frac{3\Delta'}{4\sqrt{\Delta}} - i\sigma\sqrt{\Delta}\left(\frac{K}{\Delta} - \eta\omega\right), \qquad R_1 = -\sigma\frac{2\omega Q(Mr - Q^2)}{d} + i\beta. \tag{3}$$

The constant Hermitian matrices obey

$$\mathcal{I}^2 = \mathcal{J}^2 = 1, \quad \mathcal{I}\mathcal{J} = -\mathcal{J}\mathcal{I}, \quad \{\mathcal{A}, \mathcal{I}\} = \{\mathcal{A}, \mathcal{J}\} = 0. \tag{4}$$

Thus $\mathcal{S} = \mathcal{I}\mathcal{J}$ has eigenvalues $i, -i$ and commutes with $\mathcal{A}$. On $L^2((0,\pi), d\theta; \mathbb{C}^4)$,

$$\mathcal{A}_\sigma = \begin{pmatrix} i\partial_\theta & i\Theta \\ -i\Theta & -i\partial_\theta \end{pmatrix}, \qquad \Theta = \begin{pmatrix} W & T \\ T & -W \end{pmatrix}, \tag{5}$$

$$W = \frac{3}{2}\cot\theta + \sigma m \csc\theta - \sigma a\kappa\omega\sin\theta, \qquad T = -\beta - \sigma ac\omega\cos\theta.$$

One may take $\mathcal{I} = \left(\begin{smallmatrix} 0 & I_2 \\ I_2 & 0 \end{smallmatrix}\right)$ and $\mathcal{J} = \left(\begin{smallmatrix} 0 & -J_2 \\ J_2 & 0 \end{smallmatrix}\right)$, where $J_2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. All relations are complex algebraic identities; no conjugation of ω is used.

2.2 Invertibility and boundary fidelity

Write $\rho = r + ia\cos\theta$, $\rho^* = r - ia\cos\theta$, and

$$X = \begin{pmatrix} Q(\sqrt{\Delta} + a\sin\theta)/\rho & -i(M - Q^2/\rho) \\ i(M - Q^2/\rho^*) & Q(\sqrt{\Delta} - a\sin\theta)/\rho^* \end{pmatrix}. \tag{6}$$

Its determinant is $-d \neq 0$. The source transformation is

$$y_\pm^{(+)} = \Delta^{5/4}\sqrt{\sin\theta}\begin{pmatrix} \psi_2 \pm i\Delta^{-1/2}\psi_1 \\ -2Q(k \pm i\Delta^{-1/2}s) \end{pmatrix},$$

$$y_\pm^{(-)} = \Delta^{-3/4}\sqrt{\sin\theta}\begin{pmatrix} \psi_{-2} \mp i\Delta^{1/2}\psi_{-1} \\ 2Q(n \mp i\Delta^{1/2}\ell) \end{pmatrix},$$

and $Z = (z_+, z_-)^t$, where $z_+ = \mathrm{adj}(X)y_+$, $z_- = X^* y_-$. The matrix X depends on the real background and coordinates, not frequency. All factors are invertible in the open exterior. The substitutions in the original first-order equations give (2) without dividing by any coefficient of a second-order elimination.

Let $\Pi_\pm = (1 \pm \mathcal{I})/2$. The source transformation gives

	$\Pi_+ Z$	$\Pi_- Z$
$\sigma = +1$	$H\Delta^{-1/4}C^\infty$	$H\Delta^{-3/4}C^\infty$
$\sigma = -1$	$H\Delta^{3/4}C^\infty$	$H\Delta^{5/4}C^\infty$.

(7)

Smoothness may be interpreted into the angular Hilbert space. Physical axis regularity implies this condition and the maximal angular domain at each interior radius. At infinity the transformed components are H^{-1} times polynomially bounded functions; the remaining transformation factors have polynomial bounds.

For fixed $Q \neq 0$, the phantom tetrad gauge is legitimate because the background Maxwell component $\Phi_1 = Q/(2\rho^{*2})$ is nowhere zero in the exterior. The radiative variables are the rescaled phantom-gauge Weyl components

$$\psi_2 = \Psi_0^{(1)}, \quad \psi_1 = \sqrt{2}\rho^{*}\Psi_1^{(1)}, \quad \psi_{-1} = \rho^{*3}\Psi_3^{(1)}/\sqrt{2}, \quad \psi_{-2} = \rho^{*4}\Psi_4^{(1)}. \tag{8}$$

The exact radiative data meant in the theorem are $\mathcal{G}_\pm, \mathcal{E}_\pm$ of [M, equations (5.1)–(5.2)]:

$$\begin{aligned} \mathcal{G}_+ &= \delta\Psi_0, & \mathcal{G}_- &= \rho^{*4}\delta\Psi_4, \\ \mathcal{E}_+ &= \frac{\rho^{*}}{3M}(Q\delta\Psi_1 - 3\rho^{*2}\Psi_2^{(0)}\delta\Phi_0), \\ \mathcal{E}_- &= \frac{\rho^{*3}}{3M}(Q\delta\Psi_3 - 3\rho^{*2}\Psi_2^{(0)}\delta\Phi_2). \end{aligned} \tag{8a}$$

The mixed combinations are invariant under the infinitesimal null rotations used to impose phantom gauge. In that gauge,

$$\mathcal{G}_\pm = \psi_{\pm 2}, \qquad \mathcal{E}_+ = \frac{Q}{3\sqrt{2}M}\psi_1, \qquad \mathcal{E}_- = \frac{\sqrt{2}Q}{3M}\psi_{-1}. \tag{8b}$$

Thus proving the four ψ's vanish proves vanishing of these well-defined radiative data. At finite charge, $\mathcal{E}_\pm$ contain both curvature and Maxwell perturbations. Their zero-charge limits in a nonsingular tetrad are $\delta\Phi_0$ and $\rho^{*2}\delta\Phi_2$, respectively. These local algebraic identifications do not require a real-frequency flux formula. We use the one-way implication from physical perturbations to the necessary first-order systems. Reconstruction of arbitrary subsystem solutions is not an assumption.

3 Complex angular analysis

3.1 Fixed domain and exact zero-frequency spectrum

On $\ker(\mathcal{S} - i)$, a constant unitary identification gives

$$B_\sigma(\omega) = B_0 + \omega V, \qquad B_0 = i\sigma_3\partial_\theta + g\sigma_2 - \beta\sigma_1, \qquad g = \frac{3}{2}\cot\theta + \sigma m \csc\theta, \tag{9}$$

$$V = -\sigma a(\kappa \sin\theta\, \sigma_2 + c\cos\theta\, \sigma_1).$$

Here σ_j are the Hermitian Pauli matrices; the scalar system sign σ is distinct.

The compactly supported smooth realization of B_0 is essentially self-adjoint, with maximal distributional L^2 domain. This follows also from [H, Lemma 1] at zero frequency. The endpoint indicial roots are $\pm(3\cos\theta_0/2 + \sigma m)$, nonzero half-integers. The root at or below $-1/2$ is not square integrable. The deficiency integration has zero boundary flux and forces every deficiency vector to vanish. The bounded Hermitian multiplier V leaves the operator domain fixed at complex ω.

A constant Hadamard rotation puts B_0 in the form

$$\begin{pmatrix} -\beta & i(\partial_\theta + g) \\ i(\partial_\theta - g) & \beta \end{pmatrix}. \tag{10}$$

Its square minus β^2 has diagonal entries

$$L_s = -\partial_\theta^2 + \frac{m^2 + s^2 - \frac{1}{4} + 2\sigma m s\cos\theta}{\sin^2\theta} - \frac{9}{4}, \qquad s = 1, 2. \tag{11}$$

Let $A_s = |\sigma m + s|$, $B_s = |\sigma m - s|$. The functions

$$\sqrt{\sin\theta}(1 - \cos\theta)^{A_s/2}(1 + \cos\theta)^{B_s/2}P_n^{(A_s,B_s)}(\cos\theta), \qquad n \geq 0, \tag{12}$$

have eigenvalue $\ell(\ell + 1) - 2$, where $\ell = n + \max(|m|, s)$, by the Jacobi differential equation. The change of variable $x = \cos\theta$ identifies their norms with Jacobi weighted norms, so polynomial density proves completeness. At

critical exponent $1/2$, the leading coefficient of a potentially nonintegrable first-order image cancels. The displayed functions and their first-order images belong to the maximal domain. Thus (11) is precisely the square of (10), with no unauthorized logarithmic second-order boundary extension.

For $\ell \geq \max(|m|, 2)$, the off-diagonal part of (10) couples the two one-dimensional eigenspaces in (12), giving the complete paired spectrum

$$\nu_{\ell,\pm} = \pm\sqrt{\beta^2 + (\ell - 1)(\ell + 2)}. \tag{13}$$

If $|m| \leq 1$, there is additionally one unpaired eigenvalue $\nu = -\beta$, with upper component proportional to $(\sin\theta)^{3/2} \tan(\theta/2)^{\sigma m}$ and lower component zero. There is no unpaired $+\beta$ eigenvalue in this chiral sector. Let P_0 be that possible rank-one projection. In quadratic-form sense on $\mathrm{Dom}\, B_0$,

$$T_0 := B_0^2 - \beta^2 \geq 4(1 - P_0) \geq 0. \tag{14}$$

3.2 The required nonnegative angular form

For real $x \leq -h$, set

$$K_\sigma(x) = B_0^2 - \beta^2 + \frac{c}{x}(B_0 - \sigma\beta). \tag{15}$$

This form is nonnegative for both signs of Q and σ. Indeed, put $q = |Q|/M \in (0, 1)$, $b = |\beta| = 3/(2q)$. Then $|c/x| \leq q$, with sign opposite to β. For $\sigma = -1$, the smaller nonzero-branch value is bounded by

$$v^2 - b^2 - q(v + b), \qquad v \geq \sqrt{b^2 + 4}.$$

It increases with v, and its endpoint value is

$$\epsilon_q = \frac{5 - \sqrt{9 + 16q^2}}{2} = \frac{8(1 - q^2)}{5 + \sqrt{9 + 16q^2}} > 0.$$

The unpaired branch has value zero. Hence

$$K_-(x) \geq \epsilon_q(1 - P_0). \tag{16}$$

For $\sigma = +1$, the smaller nonzero-branch value is at least $(11 - \sqrt{9 + 16q^2})/2 > 3$, and the unpaired value is $-2\beta c/x = -3h/x > 0$. Consequently

$$K_+(x) \geq -\frac{3h}{x}P_0 + \frac{11 - \sqrt{9 + 16q^2}}{2}(1 - P_0) \geq 0. \tag{17}$$

No uniform estimate at the extremal boundary $q = 1$ is used.

For a normalized genuine eigenvector $B_\sigma(\omega)u = \lambda u$, introduce real numbers

$$L = \langle u, B_0 u\rangle, \quad v = \langle u, Vu\rangle, \quad T = \|B_0 u\|^2 - \beta^2,$$
$$C = 2\,\mathrm{Re}\langle B_0 u, Vu\rangle, \quad D = \|Vu\|^2.$$

With the inner product conjugate-linear in its first entry,

$$\lambda = L + \omega v, \qquad \lambda^2 = \beta^2 + T + \omega C + \omega^2 D. \tag{18}$$

The latter identity is valid using only the first-order domain:

$$\langle B_\sigma(\omega)^* u, B_\sigma(\omega)u\rangle = \lambda\langle B_\sigma(\omega)^* u, u\rangle = \lambda\langle u, B_\sigma(\omega)u\rangle = \lambda^2.$$

Expanding the left side proves (18). This does not presume self-adjointness at complex frequency or a square-operator domain. The Pauli algebra gives

$$V^2 = a^2(c^2 + \sin^2\theta)1, \qquad \|V\| = |a|\kappa. \tag{19}$$

The eigenvector in these expectations is fixed independently of the transform coordinate x.

3.3 Generalized-root completeness

For each fixed m, (13) implies that one resolvent of B_0, and of the full $\mathcal{A}_0$, belongs to Schatten class $\mathcal{S}_p$ for every $p > 1$, in particular $\mathcal{S}_2$. The bounded perturbation preserves this property at a sufficiently distant resolvent point. If $C_\omega = |\omega|\|\mathcal{V}\|$, where $\mathcal{V}$ is the full-space frequency multiplier,

$$\|(\mathcal{A}(\omega) - z)^{-1}\| \le \frac{1}{|\operatorname{Im} z| - C_\omega} \quad \text{when } |\operatorname{Im} z| > C_\omega. \tag{20}$$

The resolvent identity at $z = i(1 + 2C_\omega)$ gives an actual Hilbert–Schmidt resolvent.

The general theorem in Mityagin–Siegl [MS, Theorem 2.2] says that a closed densely defined operator with one $\mathcal{S}_p$ resolvent has complete generalized root spaces if its resolvent has polynomial bounds on rays with successive gaps strictly less than π/p. Take $p = 2$ and the eight nonreal rays $\pi/8 + j\pi/4$, $j = 0, \dots, 7$. Equation (20) gives the required bounds at large radius; the gap is $\pi/4 < \pi/2$. Thus the hypotheses of the completeness theorem hold. No Riesz-basis property is needed.

The same argument applies to $\mathcal{A}(\omega)^* = \mathcal{A}(\overline{\omega})$. Therefore, if every finite Riesz projection annihilates a vector, that vector is zero, because it is orthogonal to the complete generalized root spaces of the adjoint. This detection statement replaces any presumed convergent expansion in ordinary eigenvectors.

4 Full PDE reduction at defective angular frequencies

Let P^+_λ be a Riesz projection in the i chiral sector. On the two-sector direct sum use

$$\mathcal{P}_\lambda = \operatorname{diag}(P^+_\lambda, \mathcal{I} P^+_\lambda \mathcal{I}). \tag{21}$$

It commutes with $\mathcal{A}, \mathcal{I}, \mathcal{J}$ by their resolvent identities. This refined pairing avoids double counting if both $\lambda, -\lambda$ occur in one sector and works at $\lambda = 0$. Adjoint-root completeness detects any nonzero full PDE solution at some interior radius with such a nonzero finite projection.

Choose Jordan chains $(\mathcal{A} - \lambda)u_j = u_{j-1}$, $u_0 = 0$, in the i sector; set $v_j = \mathcal{I} u_j$. Then

$$\mathcal{A} v_j = -\lambda v_j - v_{j-1}, \quad \mathcal{I} u_j = v_j, \quad \mathcal{I} v_j = u_j, \quad \mathcal{J} u_j = i v_j, \quad \mathcal{J} v_j = -i u_j. \tag{22}$$

The two chiral chains remain independent when $\lambda = 0$. If their radial coefficients are p_j, q_j, equation (2) reads

$$\sqrt{\Delta}\, p_j' = \lambda p_j + p_{j+1} + (R_0 - iR_1) q_j, \qquad \sqrt{\Delta}\, q_j' = -\lambda q_j - q_{j+1} + (R_0 + iR_1) p_j. \tag{23}$$

Choose a chain and its highest index whose pair is not identically zero. Higher coefficients vanish identically, leaving a homogeneous ordinary two-component radial equation. The coefficient map treats $u_j, \mathcal{I} u_j$ in the same way and intertwines $\mathcal{I}, \mathcal{J}$. The finite projections and coefficient maps are bounded and independent of radius. They preserve the different horizon weights in (7) and the infinity bounds.

Equivalently, use the finite invariant filtration $\ker(B - \lambda)^j \oplus \mathcal{I} \ker(B - \lambda)^j$. The radial equation preserves it, and the first nonzero quotient has semisimple angular action. This version does not apply an anticommuting angular operator to the entire PDE.

In the radial basis $Z = z_1(u + \mathcal{I} u) - i z_2 (u - \mathcal{I} u)$, the pair is

$$(\partial_r \mp \Delta^{-1/2} R_0) z_{1/2} = \Delta^{-1/2} (R_1 \mp i\lambda) z_{2/1}. \tag{24}$$

The two horizon powers are $(-1/4, -3/4)$ for positive spin, $(3/4, 5/4)$ for negative spin, multiplied by H. It suffices to exclude (24) for every genuine angular eigenpair. No uniform estimate in m, ω, λ has been assumed.

5 A radial equation without scalar division

Define

	w	$w_\sharp$
$\sigma = +1$	$\Delta^{3/4} H^{-1} e^{i\eta\omega r} z_2$	$\Delta^{1/4} H^{-1} e^{i\eta\omega r} z_1$
$\sigma = -1$	$\Delta^{-3/4} H^{-1} e^{i\eta\omega r} z_1$	$\Delta^{-5/4} H^{-1} e^{i\eta\omega r} z_2$.

(25)

Both functions are smooth at the horizon, and

$$w' = Aw_\sharp, \qquad \Delta w'_\sharp + \left(\frac{1-3\sigma}{2}\Delta' + 2i\widetilde{K}\right) w_\sharp = Bw, \tag{26}$$

where $A = R_1 + i\sigma\lambda$, $B = R_1 - i\sigma\lambda$. Put

$$p = \Delta^{-(1+3\sigma)/2} H^2 e^{-2i\eta\omega r}, \qquad v_r = pw, \qquad z_r = \Delta p w_\sharp. \tag{27}$$

The subscripts distinguish these radial functions from angular expectations. Direct differentiation gives

$$(\Delta\partial_r + L_r)v_r = Az_r, \qquad z'_r = Bv_r, \qquad L_r = \frac{1+3\sigma}{2}\Delta' - 2i\widetilde{K}. \tag{28}$$

Since $A' = B' = R'_1$ is constant, two derivatives of the first equation, using the second, give

$$\mathcal{E}v_r = 0, \qquad \mathcal{E} = \partial_r^2(\Delta\partial_r + L_r) - (R_1^2 + \lambda^2)\partial_r - (3R_1 - i\sigma\lambda)R'_1. \tag{29}$$

All coefficients are polynomials of degree at most two in r. Neither A^{-1} nor B^{-1} is used. Apparent scalar poles therefore cause no loss of solutions.

6 Distributional horizon extension

Write $t = r - r_+$, $d_H = \Delta'(r_+) = 2\delta$. Locally

$$v_r = t^\alpha f(t), \quad z_r = t^{\alpha+1} g(t), \quad f, g \in C^\infty, \qquad \alpha = -\frac{1+3\sigma}{2} + \frac{2iK(r_+)}{d_H}. \tag{30}$$

Negative spin has $\operatorname{Re}\alpha > 1$, positive spin has $\operatorname{Re}\alpha > -2$, since $\operatorname{Re}(2iK(r_+)/d_H) = 2(r_+^2 + a^2)\gamma/d_H$.

6.1 Generic frequencies

For negative spin ordinary zero extension solves (28) distributionally. For positive spin with $\alpha \neq -1$, use the canonical meromorphic distributions t_+^α and $t_+^{\alpha+1}$, multiplied by f, g, and joined to the original solution away from the horizon. Their only pole in the stated region is $\alpha = -1$. The identities

$$t\partial_t(t_+^\alpha f) = \alpha t_+^\alpha f + t_+^{\alpha+1} f', \qquad \partial_t(t_+^{\alpha+1} g) = (\alpha+1)t_+^\alpha g + t_+^{\alpha+1} g' \tag{31}$$

hold by meromorphic continuation from the integrable region, including at nonnegative integers. With $\Delta = t(d_H + t)$, the residuals in (28) are the canonical extensions of identically vanishing smooth coefficient residuals. Thus the extended pair is homogeneous. This is an argument at fixed ω, requiring no holomorphic family of physical outgoing solutions.

6.2 The exceptional positive-spin resonance

The only exceptional negative integer is $\alpha = -1$, at

$$\omega = \omega_H = \frac{am + id_H/2}{B_+}, \qquad B_+ = r_+^2 + a^2, \qquad \gamma = \frac{\delta}{B_+}. \tag{32}$$

At this point $L_r(r_+) = d_H$. Define $G_H = 2Q(Mr_+ - Q^2)/d$. If $B(r_+) = 0$, then $\lambda = \beta + iG_H\omega$, so $\lambda^2 - \beta^2 = 2i\beta G_H\omega - G_H^2\omega^2$. Compare with (18), divide by $\omega \neq 0$, and take imaginary parts:

$$0 = -\frac{\gamma T}{|\omega|^2} + \gamma(D + G_H^2) - 2\beta G_H. \tag{33}$$

But $T \geq 0$, $D \leq a^2\kappa^2$, and

$$G_H^2 + a^2\kappa^2 = a^2 + \frac{4Q^2 B_+}{d},$$

so

$$2\beta G_H - \gamma(G_H^2 + a^2\kappa^2) = 6M + \delta\left(\frac{6M^2 - 4Q^2}{d} - \frac{a^2}{B_+}\right) > 0. \tag{34}$$

The first quotient in parentheses is $4 + 2M^2/d > 6$, while $a^2/B_+ < 1$. Both signs of Q are included. This contradicts (33), so $B(r_+) \neq 0$. The same squared argument would exclude $A(r_+) = 0$, but that is unnecessary below.

Here z_r is smooth and $v_r = t^{-1}f$. The equation $z_r' = Bv_r$ forces $f(0) = 0$, so v_r is also smooth. Its zero extension need not be homogeneous: z_r' acquires a delta source. The required corrected extension is

$$z_e = 1_{[r_+,\infty)} z_r, \qquad v_e = 1_{[r_+,\infty)} v_r + \frac{z_r(r_+)}{B(r_+)} \delta_{r_+}. \tag{35}$$

Since

$$(\Delta\partial_r + L_r)\delta_{r_+} = (-d_H + L_r(r_+))\delta_{r_+} = 0,$$

the first equation of (28) is unchanged. The added term in Bv_e precisely matches the delta coefficient in z_e'. Thus (28), and hence (29), hold homogeneously.

This correction is essential: vanishing of the Laurent coefficient alone does not remove a finite horizon boundary source. The corrected extension is of finite distribution order, supported in $[r_+, \infty)$, and agrees with the physical function on the open exterior.

7 Complex Laplace transform and positive energy

7.1 Transform equation

At infinity the homogeneous extension v_e agrees with $e^{-i\eta\omega r} O_{\mathrm{con}}(r^C)$, by (25)–(27). Its Fourier–Laplace transform

$$F(\xi) = \langle v_e(r), e^{-i\xi r} \rangle \tag{36}$$

is holomorphic for $\operatorname{Im}\xi < -\eta\gamma$. The pairing is defined by exponential damping and cutoffs; compactly supported finite-part or delta terms present no convergence problem. Restrict to the interior ray

$$\xi = \omega x, \qquad -\infty < x < -h, \qquad Y(x) = F(\omega x). \tag{37}$$

Here $\gamma(x + \eta) < -2\gamma$.

Transform (29) by $\partial_r \mapsto i\xi$, $r \mapsto i\partial_\xi$ in the correct operator order, then set $\xi = \omega x$. The result is

$$a_x Y'' + (b_x + i\omega c_x)Y' + (d_0 + \omega d_1 + \omega^2 d_2 + i\omega e_x)Y = 0, \tag{38}$$

where

$$\begin{aligned} a_x &= x(x+\eta)(x+h), & b_x &= x^2 - \eta h - 3\sigma x(x+h), \\ c_x &= 2Mx(x+\eta)^2, & e_x &= M(x+\eta)(x-\eta-3\sigma x), \\ d_0 &= (\beta^2 - \lambda^2)x + \tfrac{3}{2}\sigma h - c\lambda, & d_1 &= 2amx^2, \\ d_2 &= -[2a^2 + (a^2+Q^2)(x+2\eta)]x^2 - Q^2\eta^2 x. \end{aligned} \tag{39}$$

The coefficients in (39) follow by applying the distributional Fourier substitution to the polynomial radial operator (29), preserving the order of multiplication and differentiation.

Set

$$\mu = x^{-2}(x+\eta)^{-3\sigma}, \quad P = \mu a_x = \frac{x+h}{x(x+\eta)^{3\sigma-1}}, \quad W_x = \frac{(x+\eta)^{-3\sigma}}{x}. \tag{40}$$

Both P, W_x are strictly positive on this interval. With $q_x = \mu c_x/2 = M(x+\eta)^{2-3\sigma}/x$,

$$(\mu a_x)' = \mu b_x, \qquad q_x' = \mu e_x, \qquad q_x/P = \frac{M(x+\eta)}{x+h}. \tag{41}$$

Define

$$S(x) = M[x - 2\log(-x-h)], \qquad Y = e^{-i\omega S} U. \tag{42}$$

Because $S' = q_x/P$, the imaginary transport disappears:

$$(PU')' + W_x \left[\beta^2 - \lambda^2 + \frac{3\sigma h/2 - c\lambda}{x} + 2am\omega x + \omega^2 R(x)\right] U = 0, \tag{43}$$

$$R(x) = (x+\eta)^2 \left(\delta^2 - \frac{2M^2}{x+h}\right) + a^2\eta^2 - 2a^2 x. \tag{44}$$

7.2 The coercive sign

Use the genuine angular eigenvector in (18). The bracket in (43) equals

$$-k_\sigma(x) + \omega l(x) + \omega^2 \mathcal{B}(x), \tag{45}$$

where

$$k_\sigma = T + \frac{c}{x}(L - \sigma\beta) = \langle K_\sigma(x) \rangle \geq 0, \qquad l(x) = 2amx - C - \frac{c}{x}v \in \mathbb{R}, \qquad \mathcal{B} = R - D.$$

For $y = -x - \eta > 2$, equation (19) gives the exact identity

$$\mathcal{B}(x) = y^2 \left(\delta^2 + \frac{2M^2}{y-2} \right) + 2a^2 y - a^2 \langle \sin^2 \theta \rangle > 0. \tag{46}$$

These signs hold for every allowed background and actual complex angular eigenpair. No complex separation constant is treated as an arbitrary real parameter. The real, possibly indefinite coefficient l will cancel in the imaginary-part identity.

7.3 Endpoint justification

At $x \uparrow -h$, the Laplace transform is analytic across the endpoint, which is strictly inside its holomorphy half-plane. Writing $t_x = -x - h$, equation (42) gives

$$|U| = O(t_x^{2M\gamma}), \qquad |U'| = O(t_x^{2M\gamma - 1}). \tag{47}$$

Here $P \asymp t_x$, $W_x \asymp 1$, $\mathcal{B} \asymp t_x^{-1}$, and k_σ, l are bounded. The worst energy integrands behave as $t_x^{4M\gamma-1}$, which is integrable for every $\gamma > 0$. The flux $PU'\overline{U}$ tends to zero as $O(t_x^{4M\gamma})$.

At $x \to -\infty$, an explicit finite-part representation gives a polynomial bound times $e^{\gamma r_+ x}$ for $F(\omega x)$ and each fixed derivative. Cut off near the horizon and absorb the cutoff into the smooth f. For positive spin at a generic frequency, use

$$t_+^\alpha f = \frac{\partial_t (t_+^{\alpha+1} f)}{\alpha + 1} - \frac{t_+^{\alpha+1} f'}{\alpha + 1}. \tag{48a}$$

Both undifferentiated distributions on the right are integrable since $\operatorname{Re}\alpha > -2$. Their absolute Laplace integrals are bounded by

$$e^{\gamma r_+ x} \int_0^\varepsilon t^{\operatorname{Re}\alpha + 1} e^{\gamma x t}\, dt = e^{\gamma r_+ x} O(|x|^{-\operatorname{Re}\alpha - 2}).$$

The distributional derivative contributes at most one factor of $|x|$. Negative spin is integrable directly; the exceptional extension is an ordinary smooth zero extension plus a delta. Inserting r^j handles each derivative in x by the same argument. The smooth outgoing tail begins a positive distance from the horizon and has a stronger exponential bound for sufficiently large negative x. Thus no general support-edge distribution bound or Watson asymptotic coefficient is being presumed.

The gauge factor has modulus $e^{-\gamma M x}$ times a power of $-x - h$. Thus, after adjusting constants,

$$|U| + |U'| \leq C(1 + |x|)^N e^{\gamma(r_+ - M)x} = C(1 + |x|)^N e^{\gamma \delta x}. \tag{48}$$

Because $\delta > 0$, this decays exponentially. All weights have at most polynomial growth at this endpoint. The energy integrals converge and the boundary flux is zero, including for the delta correction (35).

7.4 Energy identity and injectivity

Multiply (43) by $\overline{U}$, integrate, and use the established endpoint bounds:

$$-\int P|U'|^2 - \int W_x k_\sigma |U|^2 + \omega \int W_x l |U|^2 + \omega^2 \int W_x \mathcal{B} |U|^2 = 0.$$

Multiplication by $\overline{\omega}$ and taking imaginary parts eliminates the real linear coefficient:

$$0 = \gamma \left[\int P|U'|^2 + \int W_x k_\sigma |U|^2 + |\omega|^2 \int W_x \mathcal{B} |U|^2 \right]. \tag{49}$$

Since $\gamma > 0$, $\omega \neq 0$, $P, W_x > 0$, $k_\sigma \geq 0$, and $\mathcal{B} > 0$, this forces $U = 0$.

The transform F vanishes on an interior complex ray, hence on its entire half-plane by the identity theorem. On any sufficiently low horizontal line it is the Fourier transform of an exponentially damped tempered distribution. Fourier injectivity gives $v_e = 0$, in particular $v_r = 0$ in the open exterior. Since $p \neq 0$, $w = 0$; (26) then gives $Aw_\sharp = 0$. The affine polynomial A is not identically zero because $Q\omega M/d \neq 0$. Its possible zero is isolated, so continuity gives $w_\sharp = 0$ everywhere. The invertible transformations give $z_1 = z_2 = 0$.

This contradicts the nonzero pair extracted from any nonzero full PDE solution. Both systems (2) vanish, and their invertible source transformations imply the radiative conclusion (8a)–(8b) for $Q \neq 0$.

8 The zero-charge Kerr case

For $Q = 0$, the background Maxwell field vanishes. Linearization then separates the gravitational and Maxwell radiative equations into the standard Teukolsky equations of spins $s \in \{-2, -1, 1, 2\}$. The Kerr mode-stability input used below is Whiting's upper-half-plane theorem [W] in the separated form recalled in [AMPW, Introduction and Section II.A.3, Definition 3], restricted to negative spins $s = -1, -2$: a regular-axis separated solution which is the *non-special ingoing Frobenius solution* at the horizon and outgoing at infinity must vanish. Positive spins will be reduced to negative spins by the spin-reversal lemma below. A general $e^{-i\omega t+im\phi}F(r,\theta)$ need not initially be a product, so the full-field reduction is also proved here.

On the fixed-m subspace of $L^2((0,\pi), \sin\theta\, d\theta)$, define the positive angular convention

$$T_s(\omega) = -\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{(m + s\cos\theta)^2}{\sin^2\theta} - a^2\omega^2\cos^2\theta + 2as\omega\cos\theta. \tag{50}$$

The zero-frequency realization $T_s(0)$ is the canonical self-adjoint spin-bundle operator restricted to this azimuthal subspace. Its orthonormal spin-weighted spherical harmonics have eigenvalues

$$\Lambda_\ell = \ell(\ell+1) - s^2, \qquad \ell \geq \max(|m|, |s|). \tag{51}$$

The domain imposes the regular spin-weighted branch. This distinction is essential when $m \pm s = 0$: a logarithmic scalar solution is also in $L^2(\sin\theta\, d\theta)$, but it is excluded by the canonical/Friedrichs domain. We do not use the maximal scalar interval domain. Smooth eigenfunctions and generalized eigenfunctions follow from elliptic regularity on the spin bundle after addition of the smooth potential.

For fixed ω, the last two terms of (50) are a bounded complex multiplication operator. Since (51) grows quadratically in ℓ, a fixed-m resolvent of $T_s(0)$ is trace class. The same bounded-perturbation and nonreal-ray argument used in (20), now with $p = 1$, proves generalized-root completeness for $T_s(\omega)$ and its adjoint. For example the same eight rays have gaps $\pi/4 < \pi$.

For clarity, the canonical eigenvalue Λ in (50) is not the standard radial separation constant. With $K_{\rm std} = (r^2 + a^2)\omega - am$, direct separation of the physical Teukolsky equation gives

$$\begin{aligned} 0 = \Delta^{-s}\partial_r(\Delta^{s+1}\partial_r R) + \frac{K_{\rm std}^2 - 2is(r-M)K_{\rm std}}{\Delta}R \\ + (4is\omega r + s - \Lambda - a^2\omega^2 + 2am\omega)R. \end{aligned} \tag{52}$$

Thus the standard radial constant is $\Lambda - s + a^2\omega^2 - 2am\omega$. The fixed-frequency PDE is a sum of a radial operator and $-T_s(\omega)$, with precisely this scalar shift.

8.1 Horizon branches and the spin-reversal identity

For the Kerr background set

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \qquad \kappa_H = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \qquad k = \omega - m\Omega_H.$$

In this section $H'/H = -iK_{\rm std}/\Delta$. Writing $R = H\Delta^{-s}f$, direct conjugation of (52) gives

$$E_s f = 0, \qquad E_s = \Delta\partial_r^2 + P_s\partial_r + C_s, \tag{53}$$

with

$$\begin{aligned} P_s &= (1-s)\Delta' - 2iK_{\rm std}, \\ C_s &= 2i(2s-1)\omega r - s - \Lambda - a^2\omega^2 + 2am\omega. \end{aligned}$$

Indeed, for $\psi = \partial_r \log(H\Delta^{-s}) = -(iK_{\rm std}+s\Delta')/\Delta$, the first-derivative coefficient after conjugation is $2\Delta\psi+(s+1)\Delta' = P_s$. The remaining $1/\Delta$ terms cancel against the potential in (52), using $\Delta' = 2(r-M)$ and $K'_{\rm std} = 2\omega r$, leaving the displayed C_s.

Lemma 8.1 (negative-spin horizon branch). Let $s \in \{-1,-2\}$ and $\Lambda \in \mathbb{C}$. If a nonzero solution of (52) satisfies $R \in H\Delta^{-s}C^\infty([r_+, r_+ + \varepsilon))$, then

$$R = c\, t^{-s-ik/(2\kappa_H)}(1+O(t)), \qquad t = r - r_+, \quad c \neq 0. \tag{54}$$

In particular, it is the non-special ingoing Frobenius solution required by [AMPW, Definition 3].

Proof. Write $d_H = r_+ - r_- > 0$, so $\Delta = t(d_H + t)$. The indicial roots of (53) for f are

$$0, \qquad k_s = s + \frac{ik}{\kappa_H}, \qquad \operatorname{Re} k_s = s - \frac{\gamma}{\kappa_H} < 0.$$

The root 0 admits a unique analytic solution with a freely chosen nonzero constant coefficient. The other independent solution has a nonzero t^{k_s} contribution, possibly accompanied by logarithms at integral root differences, and cannot be smooth at $t = 0$. A nonzero smooth f must therefore be a nonzero multiple of the root-0 solution. Since $H = t^{-ik/(2\kappa_H)}$ times a smooth nonvanishing factor and $\Delta^{-s} \sim d_H^{-s} t^{-s}$, (54) follows. □

Lemma 8.2 (spin reversal, including exceptional frequencies). Fix $s \in \{1,2\}$, $m \in \mathbb{Z}$, and $\operatorname{Im}\omega > 0$. Let $T_s(\omega)u = \Lambda u$ with $u \neq 0$ in the canonical angular domain, and let $R \neq 0$ solve (52). Suppose $R \in H\Delta^{-s}C^\infty([r_+,\infty))$ and HR has polynomial conormal bounds at infinity. Set $f = H^{-1}\Delta^s R$, and define

$$\widetilde{u}(\theta) = u(\pi - \theta), \qquad \widetilde{R} = H\Delta^s \partial_r^{2s} f. \tag{55}$$

Then $\widetilde{u}$ is a canonical spin-$(-s)$ angular eigenfunction with eigenvalue Λ, and $\widetilde{R}$ is a nonzero spin-$(-s)$ radial solution satisfying the same future-horizon regularity and outgoing infinity conditions. This holds at every frequency in the open upper half-plane, without a nonvanishing horizon leading-coefficient assumption.

Proof. Reflecting $\theta \mapsto \pi - \theta$ in (50) interchanges $T_s(\omega)$ and $T_{-s}(\omega)$ at the same (ω, m, Λ). It maps the regular spin-bundle domains into one another: the pole exponents $|m+s|$ and $|m-s|$ are exchanged. This proves the angular claim, including its domain statement.

For the radial equation put $n = 2s$. Because Δ and P_s are quadratic and C_s is affine in r, Leibniz's rule applied to (53) yields

$$\boxed{\partial_r^{2s} E_s = E_{-s}\partial_r^{2s}.} \tag{56}$$

To verify all coefficients, the coefficient of ∂_r^{n+1} is $n\Delta' + P_s = P_{-s}$; that of ∂_r^n is

$$\frac{n(n-1)}{2}\Delta'' + nP_s' + C_s = C_{-s};$$

and the coefficient of ∂_r^{n-1} vanishes since

$$\frac{n(n-1)}{2}P_s'' + nC_s' = 2i\omega n(2s-n) = 0.$$

There are no further lower-order terms. Thus $g = \partial_r^{2s} f$ satisfies $E_{-s}g = 0$, and the conjugation (53) for $-s$ proves that $\widetilde{R} = H\Delta^s g$ solves the spin-$(-s)$ version of (52). Since f is smooth at the horizon, so is g. This establishes $\widetilde{R} \in H\Delta^s C^\infty$, including the resonant Frobenius cases.

It remains to exclude a zero image and check infinity. For $\gamma > 0$, the two radial exponential types are distinct, and polynomial conormal control of HR selects the outgoing solution. The outgoing asymptotic expansion of (52), including differentiated expansions, gives

$$f = e^{2i\omega r} r^{4iM\omega - 1}\big(c_0 + O(r^{-1})\big), \qquad c_0 \neq 0. \tag{57}$$

Here $H = e^{-i\omega r} r^{-2iM\omega}(\text{constant} + O(r^{-1}))$. Consequently

$$\partial_r^{2s} f = (2i\omega)^{2s} e^{2i\omega r} r^{4iM\omega - 1}\big(c_0 + O(r^{-1})\big). \tag{58}$$

Since $\omega \neq 0$, this is not identically zero. Furthermore $H\widetilde{R} = H^2\Delta^s\partial_r^{2s} f$ has polynomial conormal bounds and belongs to the spin-$(-s)$ outgoing branch. Thus $\widetilde{R}$ is nonzero, with the stated boundary conditions. □

For an angular eigenfunction the usual Kerr admissibility inequality is also automatic in our normalization:

$$\operatorname{Im}(\overline{\omega}\Lambda)\|u\|^2 = -\gamma\left(\langle u, T_s(0)u\rangle + a^2|\omega|^2\|\cos\theta\, u\|^2\right) < 0, \tag{59}$$

since the canonical $T_s(0)$ has eigenvalues $\ell(\ell+1) - s^2 \geq |s| > 0$ for $s = \pm 1, \pm 2$.

8.2 Completion of the full Kerr field argument

Any nonzero full PDE field has a nonzero finite Riesz projection by adjoint-root completeness. On a Jordan chain $(T_s - \Lambda)u_j = u_{j-1}$, its highest nonzero radial coefficient solves (52) homogeneously. The associated bottom angular vector is a genuine regular eigenfunction of (50). Projections and coefficient maps are bounded and independent of r, so they preserve the scalar spin-dependent horizon factor $H\Delta^{-s}$, the smooth horizon coefficient and the outgoing infinity exponential with polynomial bounds. At infinity, the two radial asymptotic exponentials are distinct for $\gamma > 0$; the polynomial bound after multiplication by H selects the outgoing one. The resulting nonzero separated mode has a regular angular eigenfunction, weighted-smooth horizon data and outgoing infinity behavior. If $s = -1, -2$, Lemma 8.1 identifies its horizon behavior with the non-special ingoing Frobenius branch, so Whiting's negative-spin theorem excludes it. If $s = 1, 2$, Lemma 8.2 produces a nonzero separated outgoing mode of spin $-s$ satisfying the same negative-spin theorem. Either possibility is a contradiction. Descending the finite Jordan chain and invoking adjoint-root completeness, the entire fixed-frequency Kerr radiative field vanishes, including for $a = 0$.

Separation conventions. The radial equation is the physical Teukolsky equation [T], and its symmetrized operator formulation is also discussed in [PV]. The canonical angular eigenvalue Λ in (50) differs by a scalar shift from the radial separation constants common in the literature. The conversion between (50)–(52) and the AMPW normalization $\widetilde{R} = \Delta^{s/2}R$ is specified after [AMPW, equation (7)]: $\Lambda_{\mathrm{AMPW}} + 2am\omega - a^2\omega^2 = A_{\mathrm{Whiting}} + s$, where $A_{\mathrm{Whiting}} = \Lambda - s$ in our convention. Only the negative-spin, non-special form of the Kerr theorem is invoked. In particular the argument does not divide by a positive-spin horizon Teukolsky–Starobinsky coefficient or require a positive-spin horizon leading coefficient to be nonzero.

This completes the zero-charge sector, and hence the theorem. □

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